Embeddings and neighbourhoods

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Embeddings and neighbourhoods before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Training a small image model

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image tensors and channels

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Continue the fictional robot vocabulary and add WAIT as ID 5. Use the table below, chosen by the lesson author for arithmetic practice. These coordinates are not measured robot properties and were not learned from data.

Each token ID selects one row; the resulting two-component vector is its embedding in this example.

Symbol	ID	Vector
PAD	0	[0, 0]
UNK	1	[-1, -1]

Symbol	ID	Vector
GO	2	[2, 0]
TURN	3	[1, 1]
STOP	4	[0, 2]
WAIT	5	[0, 1]

The table shape is 6 by 2: six vocabulary entries and **embedding dimension** 2. The ID sequence [2, 4, 0, 0] becomes [[2, 0], [0, 2], [0, 0], [0, 0]], with shape 4 by 2. Its mask remains [1, 1, 0, 0].

The vector dimension is not the sequence length or vocabulary size.

A zero PAD vector does not remove the need for masking. Averaging all four vectors above gives [0.5, 0.5]; averaging only the two real vectors gives [1, 1].

UNK is a real position when it replaces an unknown symbol, so its vector is not automatically excluded.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Embeddings and neighbourhoods trace and record the first step you can explain without help.

Use W2: Calculate distances and a tie for Embeddings and **neighbourhood**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Embeddings and neighbourhoods explanation against the supplied facts before discussing it.

Use the worked case for Embeddings and **neighbourhood**s. Proposed explanation: “Token ID differences define embedding distances”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Find a neighbourhood independently

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

Can that mean identify which ordered pair produced it?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace the lookup shape

Trace the lookup shape. Apply the supplied Embeddings and neighbourhoods facts and keep your reasoning.

Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Calculate distances and a tie

> Calculate distances and a tie. Apply the supplied Embeddings and neighbourhoods facts and keep your reasoning.

For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Trace a trainable row

> Trace a trainable row. Apply the supplied Embeddings and neighbourhoods facts and keep your reasoning.

For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Find a neighbourhood independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Find a neighbourhood independently. Apply the supplied Embeddings and neighbourhoods facts and keep your reasoning.

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

Can that mean identify which ordered pair produced it?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Keep the lookup consistent

> Keep the lookup consistent. Apply the supplied Embeddings and neighbourhoods facts and keep your reasoning.

A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Embeddings and neighbourhoods to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled embeddings and **neighbourhood**s **evidence** sheet to Image-pattern Network Investigation. Show how you can look up a vector using a token id and preserve sequence shape.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Embeddings and neighbourhoods

> Measure embedding neighbours without turning numerical proximity into factual certainty.

> Explain the labelled step before checking the changed condition.

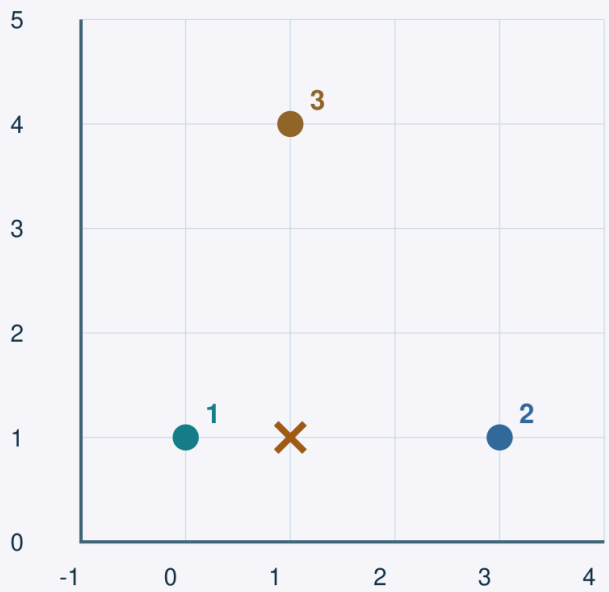
Measure embedding neighbours without turning numerical proximity into factual certainty.

Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/26

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/26

Embeddings and neighbourhoods

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance to P

(1-0)2+(1-1)2=1.



2. Distances to Q/R

Q distance 4; R distance 9.



3. Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

CHANGE ONE THING

Move the query to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Measure embedding neighbours without turning numerical proximity into factual certainty.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? P is nearest under the declared metric; this is a geometric result, not proof of meaning.

1. Distance to P

2. Select neighbour

3. Distances to Q/R

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select neighbour P is nearest under the declared metric; this is a geometric result, not proof of meaning. An embedding stores each item as an ordered vector.

Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move the query to [2, 1]. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Q becomes nearest with squared distance 1; P distance 4; R distance 10.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Changing the query position changes geometric neighbours while the stored table remains fixed.

Repair a missing building block

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Embeddings and neighbourhoods picture from its labels, then record a relevant limit.

01. Embeddings and neighbourhoods: the case

AI ACADEMY / AI-5/26 · 01 · LOOK AT THE FACTS

Embeddings and neighbourhoods: the case

The situation

Measure embedding neighbours without turning numerical proximity into factual certainty.

Supplied evidence

Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use squared Euclidean distance. All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain embeddings and neighbourhoods using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

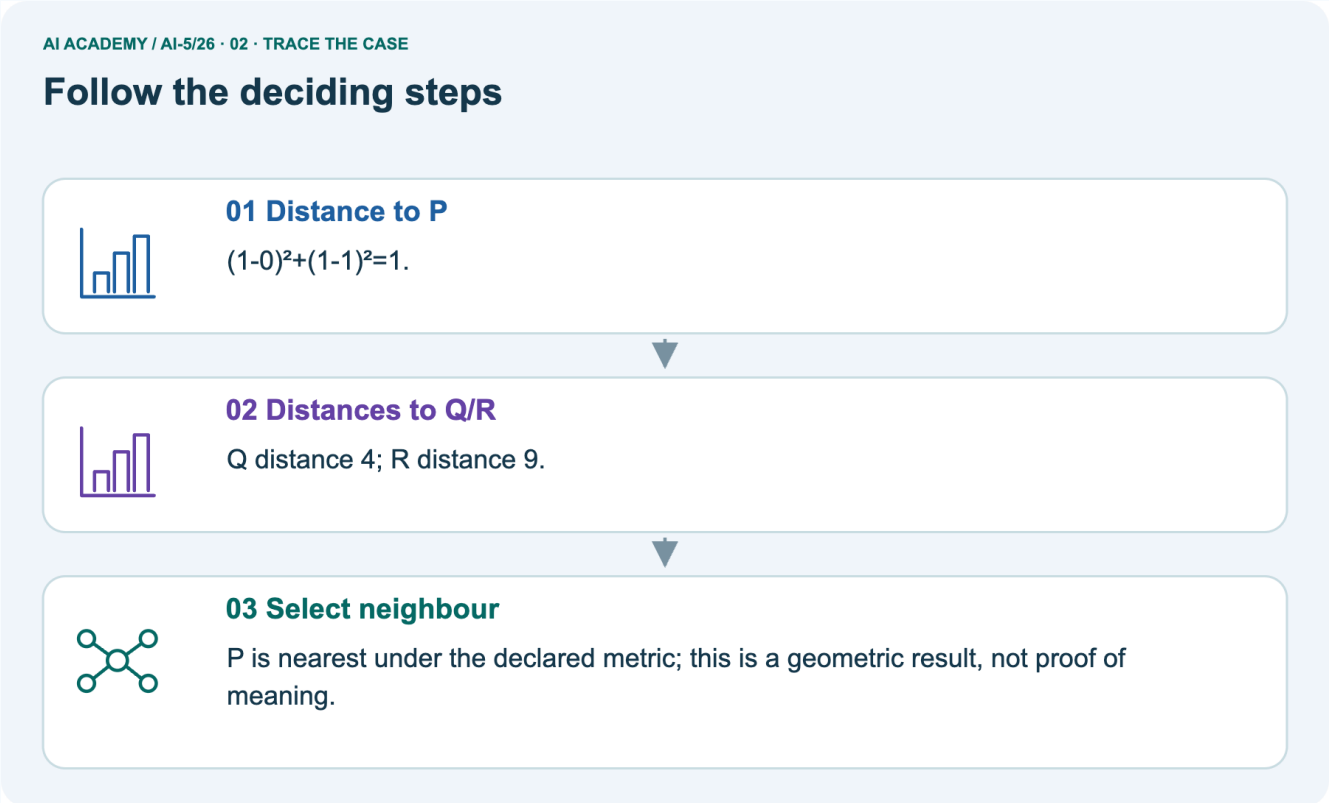
Measure embedding neighbours without turning numerical proximity into factual certainty. Supplied facts: Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Distance to P: $(1-0)^2+(1-1)^2=1$. → Distances to Q/R: Q distance 4; R distance 9. → Select neighbour: P is nearest under the declared metric; this is a geometric result, not proof of meaning.

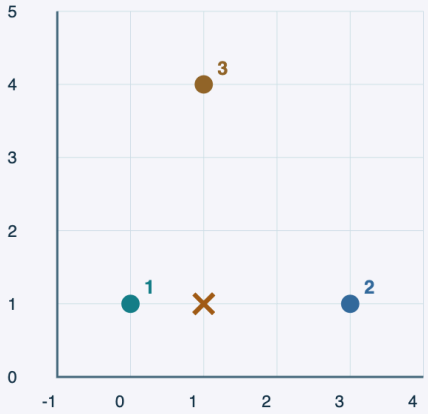
Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Squared Euclidean embedding distances

AI ACADEMY / AI-5/26

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

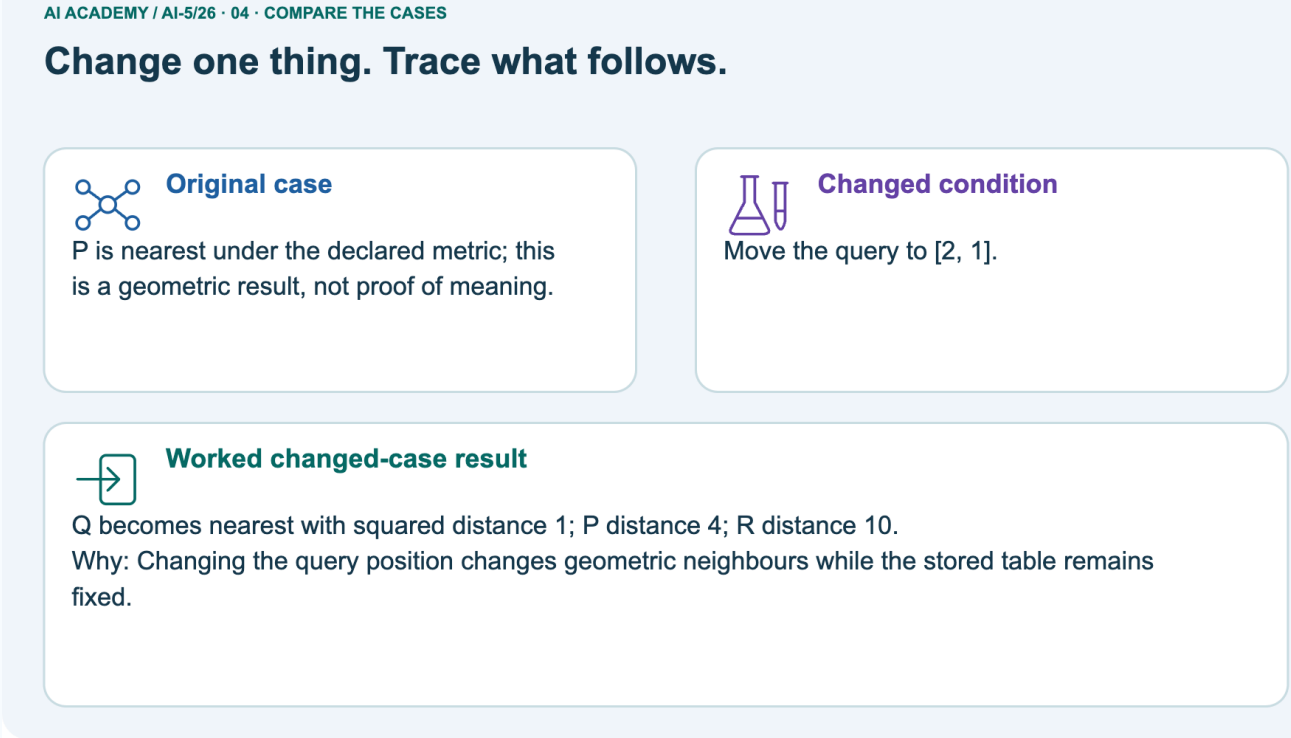
Follow the labelled picture with the matching explanation. Open the vector diagram

Squared Euclidean embedding distances. P: (0, 1). Q: (3, 1). R: (1, 4). Query query: (1, 1). Axes: first embedding coordinate; second embedding coordinate. Distance: **Squared Euclidean distance**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

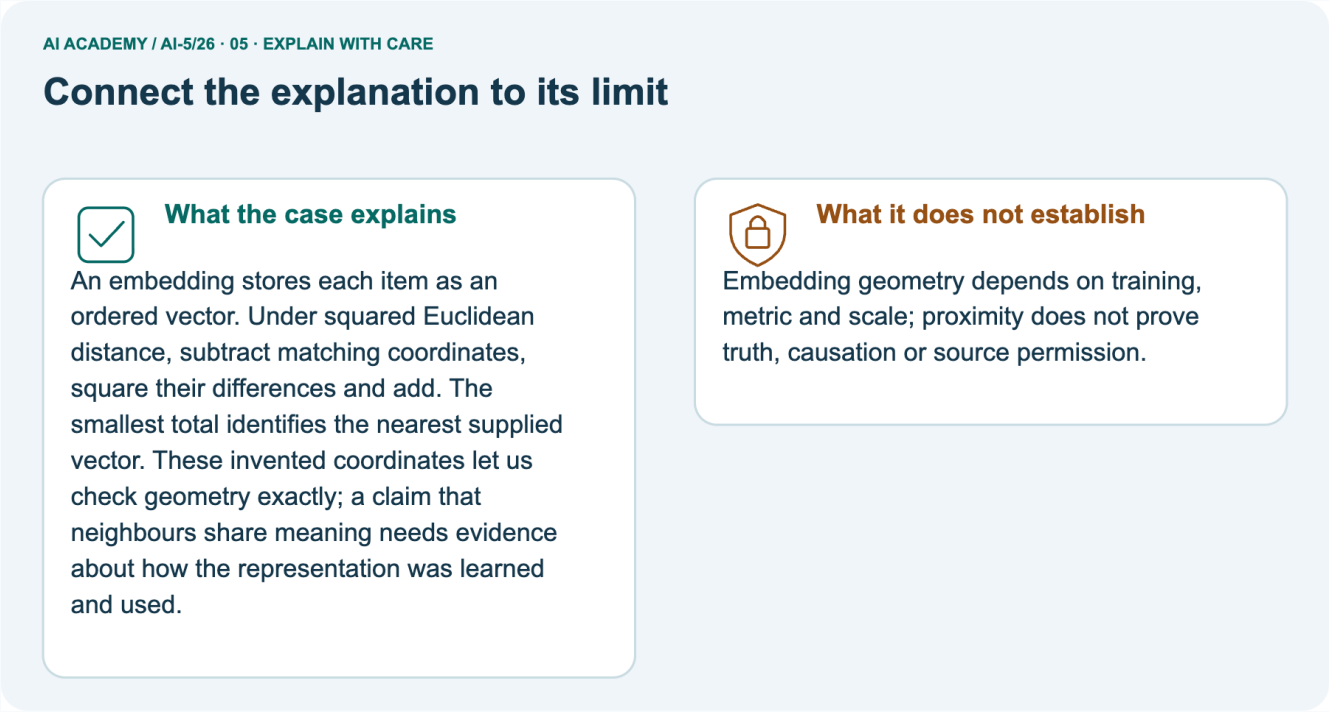
Original case: P is nearest under the declared metric; this is a geometric result, not proof of meaning. Changed condition: Move the query to [2, 1].

Expected result: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

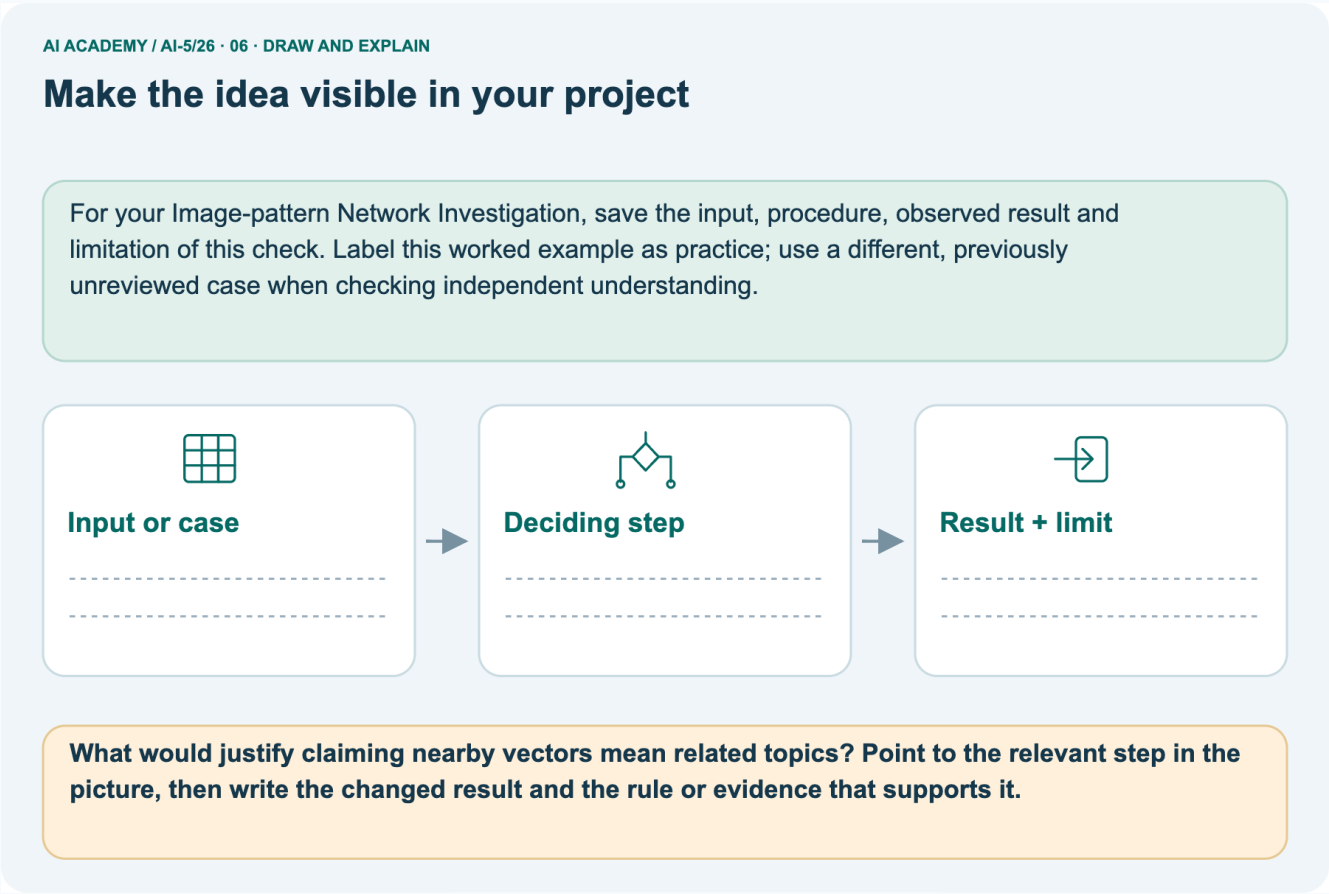
These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Limit: Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.